

A Review on various IOT-based Cattle health monitoring Systems

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Abstract

The IOT based cattle health monitoring system which utilizes Internet of Things (IoT) technology to alter the way of monitoring cattle and cared for in agricultural settings, breaking innovation in livestock management. Cattle health monitoring is essential in this modern world, because of high demand for dairy products. This paper reviews various Cattle Health Monitoring System, which leverages Internet of Things (IoT) sensors, ML, and data analytics to provide real-time insights into the health of cattle. Specially, these systems are designed to detect early signs of illness and provide data-driven decision-making tools for farmers and ranchers. Using this devices, cattle's heart rate, surrounding temperature, and location can be monitored. This IoT system is integrated with the cloud machine learning algorithm predicts the health status of cattle based on the sensor's real time data, observing the real time health status will alert the user if cattle suffer from a health issue.

Keywords: Internet of Things (IoT), Artificial Intelligence, Machine Learning

Introduction

The Internet of Things (IoT) is an emerging paradigm that bridges the gap between the physical and digital worlds to address real world challenges. IoT envisions the interconnection of a diverse range of intelligent devices and objects in our environment, which are capable of gathering, processing, and transmitting data. It has garnered significant attention from service providers, industries, and businesses, including sectors like smart grids, autonomous vehicles, healthcare, data acquisition, and smart farming. Through real time data collection, analysis, and processing, IoT is advancing the concept of smart agriculture, enhancing farm management, and enabling farmers to make more informed decisions. The growing influence of IoT in agriculture can be divided into three key areas: Precision Farming, Greenhouses, and Livestock Management. This project seeks to review and synthesize research focused on IoT-enabled livestock management. Notably, the scalability, interoperability, and open nature of IoT have revolutionized the livestock industry, benefiting both vast pastures and confined farms. Animal health can be efficiently monitored and tracked using smart wearable devices, such as collars. IoT-based livestock management goes beyond health monitoring, incorporating innovations like optimal environmental controls and field supervision for feeding strategies. Additionally, monitoring bee-hives has become an important aspect of IoT in the livestock sector. Various wireless sensors

have been employed to track behavioural patterns and detect gases.

Literature review

J. Tamilselvan M.E. and et al [1], proposed a system to monitor health and environment of cattle and to track the location of cattle. This paper highlights the health and environmental monitoring of cattle and to track location of the cattle. The values got from the various sensors are being continuously monitored in the internet. Hence it is time consuming and is also difficult to track the location of animals. In the proposed system, without human involvement health status of the animals can be monitored. This advanced cattle health monitoring system can replace this manual process for recognizing the various diseases. This system is very much helpful for farmers and also for doctors because it is accurate than manual observation.

S. D. Kuldharan and et al. [2], proposed a cow health monitoring system that highlights the importance of technology in enhancing animal health and farming practices. The system provides a cost- effective and efficient way to monitor the vital signs of cows in real-time and in enhancing animal health and farming practices. The system provides a cost- effective and efficient way to monitor the vital signs of cows in real-time and make necessary interventions to improve their health and well-being. The implementation of the time-series analysis and machine learning algorithms makes

it a robust solution for farmers and stakeholders to monitor the health of their cattle. Despite some limitations such as battery issues and server failures, the system underlines has shown significant potential in improving the quality of life for cows and the overall farming industry.

Varun Mhatre and et al. [3], proposed that the Health parameters of a cow can be tracked every minute utilizing the system proposed in this paper. Thus, abrupt transmutations in the milk yield can specifically be noted and any abnormality in the health parameters namely body temperature, relative humidity, heart rate, and rumination rate can be distinguished. Electronic components used are low-cost and hence it results in a cost-efficient system. Thus, the proposed system will essentially prove to be self-dependent with minimum user interference, unlike the existing systems which generally require constant observation which is very tedious in a subtle way.

Pawitar Dulari and et al. [4], states that data gaining unit consist of different types of biomedical sensors like body temperature sensor, blood pressure sensor, humidity sensor, heart beat rate detection sensor which is interfacing with microcontroller. The data gaining units obtain data and makes it accessible for the data administering and data interact unit (cloud, server and web page). The sensors are used for simple and general automatic measurement of numerous health factors. Such type of health sensors will be mounted on the cattle body which continuously observe the body issues of the cattle like temperature, heart beat rate etc. and deliver output in the type of electrical signal. These signals are then compared to a standard limit of normal values set as the starting point in data administering unit. If administering unit observes major changes or abnormal changes in particular cattle then they can contact to nearer animal care doctor.

K. Darvesh and et al. [5], proposed a general health monitoring system. They come to the conclusion that the discipline of animal husbandry uses an automated method for monitoring the health of cattle. Real time monitoring of a cow's general health is possible with the technique used in this paper. The system was tested on dairy cattle reliably and successfully. Since the electronic components are inexpensive, the system is cost effective. Thus, the system can monitor a cow's general health effectively.

Thisura Rajapakse and et al. [6], presented an IoT-based wearable device to monitor the health conditions and locational of cattle in large-scale farms. The proposed design is based on GSM/GPRS communication and can directly communicate the data to the IoT cloud server with mobile base stations without having an intermediate device. Therefore, the proposed wearable collar can be used in robust environments. This system has a design that considers reducing power consumption. Thus, a few power-saving strategies were applied, and evaluated their performance.

Pawitar Dulari and et al. [7], in order to observe an increasing requirement on ease of use of networks and services while maintaining privacy, and a proliferating number of "endpoints" including sensor clouds and of new net- works types, such as vehicular networks. To solve the ensuing problems, this paper has proposed using identities, more precisely virtual identities, as representation of entities of all kinds as the endpoints of communication.

Paritar Dulari and et al. [8], IoT can also be used as a useful tool for wildlife management tool. However, it can be exercised broadly at the lower possible level such as zoo first, to check its applicability and strength.

Bhatka AB and et al. [9], highlights the use of various sensors and monitoring devices, farmers can detect health issues early on and take the necessary actions to prevent diseases and other health problems. This system can also help reduce the use of antibiotics and other medication, thereby minimizing the risk of antibiotic resistance. The development of the cattle health monitoring system is still in its early stages, and there is much scope for further research and development. Additionally, there is a need for the development of user-friendly interfaces and software platforms that can facilitate easy data management and analysis. Overall, the cattle health monitoring system has the potential to revolutionize the livestock industry by improving the health and productivity of cattle while also reducing the environmental impact of livestock farming

C.H. Arora and et al. [10], discusses the integration of Internet of Things (IoT) technologies in the livestock industry has revolutionized animal management, including health monitoring, behaviour analysis, and environmental control. Real-time health monitoring through wearable sensors, smart feeding systems, and environmental sensors allows farmers to make data-driven decisions. Artificial intelligence and machine learning algorithms enable predictive analytics, disease management, and resource allocation. However, challenges like scalability, interoperability, and security need to be addressed for widespread adoption.

Dedi Darwis and et al. [11], design a system where several tools are packaged in one form, namely the Digital Smart Collar. The tool was built using a combination of Sensor Pulse Oximetry, Sensor Temperature (MLX90614), Sensor Current and voltage (INA219), Module Microcontroller (ESP 32U), and Programming Arduino 1.8.13. This technology can be used to monitor the health of large number of cattle.

Sheik Arafat I and et al. [12], proposed a system for monitoring the health of cattle in real time. It also focuses on detecting diseases at an early stage. It utilizes advanced technologies. These include Internet of Things (IoT) sensors. The study also uses LoRa connectivity and machine learning algorithms. The analysis of machine learning algorithms such as Fine Tree, Subspace kNN SVM and Weighted KNN, demonstrates their effectiveness in enhancing the

efficiency of cattle health management. By incorporating LoRa technology wireless communication becomes smooth and uninterrupted.

Pei Chen [13], proposed a system that uses the latest NB-IoT narrow-band communication means, covers a wider range, power consumption and cost are significantly reduced compared with similar products in the market. At the same time, the system is committed to achieving the lowest power consumption in terms of microprocessor selection, sensor selection, algorithm design, etc. The system also creatively improves the detection rate of estrus detection and reduces the rate of false detection through the monitoring of cow body temperature and image recognition of cow behaviour as an auxiliary method.

Jehangir Arshad and et al. [14], proposed a system that is capable of informing the user of individual cattle

health data coming from sensor nodes placed on different body areas through a secure wireless communication channel. The base station will be collecting data wirelessly from authenticated sensor nodes only and performing AI algorithms on that data after storing it in the system's database. The end-user will only be able to access information through system's web application after they have been authenticated by the health monitoring system. The user will be able to see predicted result from ANN model with performance measure evaluation of nearly 98%. The web application will be receiving signals from the base station in real-time that will be updated on the user interface. The health monitoring system will not only help the end-user to increase their production but bring the attention of others to this industry.

Ref.	Technology used	Author	Year	Observation
[1]	Arduino and IoT	J. Tamilselvan M.E, and et al.	2021	In the proposed system, without human involvement health status of the animals can be monitored. This advanced cattle health monitoring system can replace this manual process for recognizing the various diseases.
[2]	IoT	Sumit D Kuldharan, and et al.	2023	The system provides a cost- effective and efficient way to monitor the vital signs of cows in real-time and make necessary interventions to improve their health and well-being.
[3]	IoT	V a r u n Mhatre, and et al.	2020	The proposed system will essentially prove to be self-dependent with minimum user interference, unlike the existing systems which generally require constant observation which is very tedious in a subtle way.
[4]	IoT	P a w i t a r Dulari, and et al.	2023	The data gaining units obtain data and makes it accessible for the data administering and data interact unit (cloud, server and web page). The sensors are used for simple and general automatic measurement of numerous health factors.
[5]	IoT and AI	K Darvesh, and et al.	2023	Real time monitoring of a cow's general health is possible with the technique used in this paper. The system was tested on dairy cattle reliably and successfully. Since the electronic components are inexpensive.
[6]	IoT	T h i s u r a Rajapakse, and et al.	2022	The proposed wearable collar can be used in robust environments. This system has a design that considers reducing power consumption.
[7]	IoT	P a w i t a r Dulari, and et al.	2020	This paper has proposed using identities, more precisely virtual identities, as representation of entities of all kinds as the endpoints of communication
[8]	IoT	P a r i t a r Dulari, and et al.	2019	can be used as a useful tool for wildlife management tool.

[9]	I o T , ThinkSpeak , and a Mobile App	Bhatka AB, and et al.	2023	The development of the cattle health monitoring system is still in its early stages, and there is much scope for further research and development.
[10]	IoT, AI-ML	C h a r u H a r i s h Arora, and et al.	2024	Real-time health monitoring through wearable sensors, smart feeding systems, and environmental sensors allows farmers to make data-driven decisions.
[11]	IoT	D e d i Darwis, and et al.	2022	The technology built is the development of several tools that are packaged in one form, namely the Digital Smart Collar. This technology can be used to monitor the health of large number of cattle.
[12]	IoT and ML	S h e i k Arafat I, and et al.	2024	This study presents thorough framework for monitoring the health of cattle in real time. It also focuses on detecting diseases at an early stage. It utilizes advanced technologies.
[13]	N B - I o T Communication	Pei Chen	2024	This work uses the latest NB-IoT narrow-band communication means, covers a wider range, power consumption and cost are significantly reduced compared with similar products in the market.
[14]	AI and IoT	J e h a n g i r Arshad, and et al.	2023	The proposed system is capable of informing the user of individual cattle health data coming from sensor nodes placed on different body areas through a secure wireless communication channel. The user will be able to see predicted result from ANN model with performance measure evaluation of nearly 98%. The web application will be receiving signals from the base station in real-time that will be updated on the user interface.

Table 1: Various Cattle health monitoring system

Conclusion

The Cattle Health Monitoring System represents a significant advancement in the way we manage livestock health. By leveraging IoT technology, machine learning, and data analytics, this system provides farmers with real-time insights that can lead to better management practices, improved animal welfare, and increased productivity. Despite some challenges related to cost, data management, and reliability, the potential benefits far outweigh the drawbacks. As technology continues to evolve, it is likely that these systems will become more affordable and accessible to farmers of all sizes, further enhancing the sustainability and profitability of the agricultural industry.

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